

Advisor Meeting Notes (Formalized)

MatheMoria Research Planning

1. University Collaboration

Contact the Mathematics Department regarding opportunities to publish the MatheMoria app in collaboration with Loyola University. Explore how the university can support research, dissemination, and potential App Store publication.

2. Research Focus

Identify a clear, focused educational research question. Narrow the study to one key concept, such as conceptual understanding, visual learning, discovery learning, phenomenon-based learning, or game-based mathematics education.

3. Literature Review

Conduct a comprehensive review of research related to visual representations, gamification, conceptual understanding, inquiry-based instruction, and mathematics education. Use Google Scholar and Loyola University Library databases (e.g., ERIC, Education Source, and related education databases).

4. Research Design

Develop a study using MatheMoria as the instructional intervention. Potential methods include pre-tests, post-tests, classroom observations, student interviews, and gameplay experiences. Collect both quantitative and qualitative data.

5. Qualitative Analysis

Analyze interviews, student reflections, classroom observations, and emerging themes to better understand students' conceptual learning and experiences.

6. Visual Planning

Develop diagrams illustrating game mechanics, learning progressions, conceptual relationships, and the overall research framework.

7. Identify the Core Research Concept

Continue reviewing the literature to determine the primary educational theory that will support the study and guide the research framework.

8. Noyce Scholar Opportunities

Present preliminary research through the Noyce Scholarship Program. Prepare a research poster, presentation, and potentially a short published article through the Noyce program as an initial scholarly dissemination.

9. Professional Identity

Reflect on the guiding question: 'How do you see yourself in mathematics?' Use this reflection to shape your educational philosophy and long-term research agenda.

10. Institutional Review Board (IRB)

Prepare and submit an IRB application before collecting data from students to ensure the study meets ethical standards for research involving human participants.

11. Presentation Opportunities

Explore opportunities to present research at the Loyola Student Research Forum, local conferences, state conferences, national conferences, international conferences, and mathematics education organizations. Prepare a professional research poster.

12. Graduate Research Opportunities

Investigate additional opportunities for graduate student research, collaboration, and publication through Loyola University.

Immediate Next Steps

- Meet with the Mathematics Department.
- Define the central research concept.
- Begin the literature review.
- Develop a research design.
- Prepare for IRB submission.
- Plan presentations through Noyce and Loyola.